

DRONE-RL: Dynamic reinforcement learning for online navigation of UAVs in evolving environments

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ABSTRACT

Locating mobile targets in dynamic and cluttered environments, such as disaster zones or adversarial terrains, presents significant challenges due to unknown target mobility and changing environmental conditions. Unmanned Aerial Vehicles (UAVs), equipped with advanced sensing capabilities, offer a viable solution, but require adaptive planning mechanisms to navigate through non-stationary environments effectively. In this paper, we propose a hybrid learning framework for multi-target visitation that combines offline reinforcement learning (RL) and online convex optimization (OCO) to address these challenges. Specifically, we leverage Deep Deterministic Policy Gradient (DDPG) to pre-train various UAV navigation policies across representative scenarios. During deployment, an OCO-based policy selection mechanism adaptively selects the best policy in real-time that ensures responsiveness to environmental changes without retraining. Experimental results demonstrate that our approach consistently adapts to varying levels of non-stationarity and clutter, outperforming benchmark methods in adaptability and mission success. Notably, the online learner exhibits asymptotically vanishing average regret with different levels of non-stationary behaviors.

1. Introduction

The ability to locate targets is a critical task in high-stakes scenarios such as search-and-rescue, disaster response, and security missions. Locating targets, such as survivors or hostile agents, is challenging when their movement patterns are unknown. Unmanned Aerial Vehicles (UAVs) offer a promising solution, as they can operate autonomously across complex terrains. Addressing these challenges has substantial societal impact: reliable UAV path planning can accelerate the discovery of survivors.

However, these environments contain dense obstacles and exhibit significant non-stationarity statistical properties evolve over time. This includes changes in target motion, shifting obstacle configurations, targets actively evading detection, and abrupt alterations due to structural collapse or terrain deformation. For example, a UAV searching for survivors in a collapsed building must contend with shifting debris and occlusions, while survivors themselves may be moving or actively avoiding detection. These complexities motivate algorithms that remain effective

under non-stationary conditions and degrade gracefully as dynamics drift.

Traditional path-planning techniques [1,2] assume static environments or known target trajectories, which break down in dynamic settings. Reinforcement learning (RL) provides an alternative by learning search strategies under uncertainty [3]. However, RL models are trained in idealized simulations and may fail to generalize in the presence of pervasive noise and unpredictability [4]; small perturbations can induce suboptimal behavior [5]. Moreover, real-time RL deployment is computationally demanding, and most critically standard RL lacks theoretical guarantees under non-stationary or adversarial regimes, limiting reliability in mission-critical applications. Recent advances in meta-reinforcement learning [6] and transformer-based navigation [7,8] enhance generalization via attention and context inference; however, they assume stationary task distributions and do not directly address *online* non-stationarity in sensing and dynamics during cluttered, time-evolving missions.

Problem Statement: Despite significant advances, existing

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approaches struggle to address the following core challenge: *how can an autonomous UAV, equipped with limited sensing, plan and adapt its path in real time to locate multiple mobile targets in non-stationary environments?* This gap motivates adaptive learning frameworks that move beyond assumptions of static environments or purely stochastic settings, as *conventional path-planning methods assume static conditions, lack robustness to non-stationary dynamics, and fail to integrate sensing with navigation making them insufficient for real-world UAV missions that demand adaptability.*

To overcome these limitations, we leverage the *online convex optimization* (OCO) framework [9]. The OCO framework enables real-time decision-making without assuming a stationary data distribution and provides strong theoretical guarantees even under worst-case, evolving environments. Yet purely reactive OCO may be sample-inefficient during early deployment. To balance robustness and efficiency, we propose a hybrid approach: first, we pre-train a diverse set of continuous-control policies *offline* using Deep Deterministic Policy Gradient (DDPG) [10–12]. DDPG is an RL algorithm well suited for controlling continuous UAV movements. During deployment, we employ an OCO-based policy selection algorithm that adaptively switches among pre-trained policies in real time, avoiding the computational burden of online RL while enabling adaptation to non-stationary dynamics.

We also address the challenge of limited observability in cluttered environments. Prior multi-target visitation methods [3,13] rely on binary visibility, which ignores occlusions. We introduce a sensing model that integrates Through-Wall Target Detection (TWTD) [14–16]. TWTD uses radio-frequency (RF) signals to detect objects and people behind non-metallic obstacles (e.g., walls and debris). Building on recent TWTD advances [14,15] and UAV deployments [17], our model captures partial observability in cluttered environments.

The main contributions of this paper are:

- A formulation of multi-target visitation in dynamic, obstacle-rich environments with limited observability and non-stationary target motion, augmented with TWTD sensing to improve acquisition in cluttered areas.
- A hybrid framework that couples offline RL pre-training (DDPG) with an online policy-selection layer for real-time adaptation. We detail the DDPG instantiation for multi-target visitation, including a custom reward design.
- An OCO-based policy selection mechanism instantiated with EXP3, including a mini-batched variant with switching cost, together with theoretical performance guarantees.
- An empirical study is conducted to evaluate the performance of the DDPG model against Dijkstra’s algorithm across multiple scenarios with diverse clutter levels in the offline stage. We then quantify robustness under varying non-stationarity levels and compare our method with a UCB baseline and a meta-RL approach.

The remainder of the paper is structured as follows. [Section 2](#) reviews related literature. [Section 3](#) presents the system model and problem formulation. [Section 4](#) outlines the reinforcement learning strategy. [Section 5](#) describes the online policy selection mechanism. [Section 6](#) provides experimental validation and analysis.

2. Related work

The problem of UAV path planning in dynamic and uncertain environments has been extensively studied. We review key contributions across four primary dimensions: planning in unknown environments, obstacle avoidance, adaptation to non-stationary dynamics, and through-wall target detection (TWTD). Across these dimensions, we evaluate how existing approaches address mobility, sensing, and adaptivity, and identify the gaps that motivate our proposed framework. Our central departure is to *jointly* couple navigation, sensing, and online adaptation for multi-target visitation with realistic TWTD-based detection.

Planning in Unknown Environments. Path planning without prior environmental knowledge is central to autonomous UAV navigation. Methods based on Markov Decision Processes (MDPs) have been widely used for online decision-making in such contexts [18,19]. While effective in unknown terrains, formulations differ in how they model target dynamics. For example, [19] addresses maneuvering (mobile) radio-tagged objects via a jump-Markov motion model; [18] jointly performs moving-target tracking and path planning in unknown outdoor scenes; and [20] plans routes to *multiple mobile* targets with route updates as targets move. Recent survey on RL-based UAV systems [21] provide a taxonomy of planning and control tasks in unknown environments, highlighting the effectiveness of RL-based methods for navigation, trajectory planning, and resource scheduling. Multi-agent extensions have emerged, for example, the RL-based formation-control framework of [22], which handles static and dynamic obstacles in coordinated settings. However, the joint challenge of *real-time* planning and *visitation* of multiple *mobile* targets remains underexplored; our work explicitly targets this regime.

Obstacle Avoidance in Cluttered Domains. A range of algorithms has been developed for obstacle avoidance, including RL [23], potential-field methods [24], and graph-based techniques such as A* [25]. Recent work [26] combines global RRT* planning with the local Dynamic Window Approach (DWA), improving avoidance in cluttered missions. Surveys such as [21] emphasize progress in RL-based navigation and obstacle avoidance for UAVs, but also note that many methods treat motion planning separately from perception and mission-level decision making, limiting their ability to reason about sensing and safety jointly. We characterize the RL agent under different environmental configurations (target mobility and obstacle dynamics) and leverage these in the real-time online selection phase. Unlike heuristic planners that may get stuck in local minima or incur high computational costs, our RL-based approach learns complex non-linear policies that generalize across scenarios.

Game-theoretic approaches further address UAV safety: [27] introduce a finite-time safe RL control strategy for quadcopters in a multi-player nonzero-sum game. These emphasize coordination and formal guarantees. In contrast, our framework targets multi-target navigation amid dynamic obstacles with TWTD-based sensing, providing safety rooted in realistic detection models and online policy selection.

Through-Wall Target Detection (TWTD). TWTD uses modalities such as ultra-wideband radar to detect occluded targets [28,29]. It has been vital for constrained environments (e.g., search and rescue), with fidelity improved via signal processing and reconstruction [14–16]; UAV-based integration has also been investigated [17,30]. However, most studies treat detection as an isolated sensing task rather than embedding it within decision-making loops, despite the interdependence of detection strength and navigation in cluttered missions. We integrate TWTD-derived detection strength directly into policy selection, linking sensing to navigation for dynamic multi-target visitation.

Adaptation under Non-Stationary Obstacle Dynamics. Dynamics pose additional challenges. Meta-learning [31,32] and hierarchical exploration (e.g., LSP [33]) offer adaptability, yet degrade under abrupt or adversarial changes. LiDAR-based dynamics prediction [34] and temporal attention-enhanced DRL [35] improve adaptability but remain largely heuristic-driven. We adopt an adversarial bandit for online policy selection, using EXP3 to handle non-stationary rewards and to provide greater robustness than UCB in our setting. We build on [13,36], which study multi-target search with an OCO-based learner. While they consider non-stationary target mobility, they overlook dynamic obstacles and assume binary observations. We incorporate unknown dynamic obstacles and replace binary observations with TWTD-based detection strength, yielding a more realistic sensing model.

Contributions of This Work. The literature reveals three gaps: (i) most methods optimize navigation in isolation, decoupled from sensing and visitation; (ii) adaptive frameworks assume smooth dynamics and underperform in abrupt regimes; and (iii) TWTD sensing is not embed-

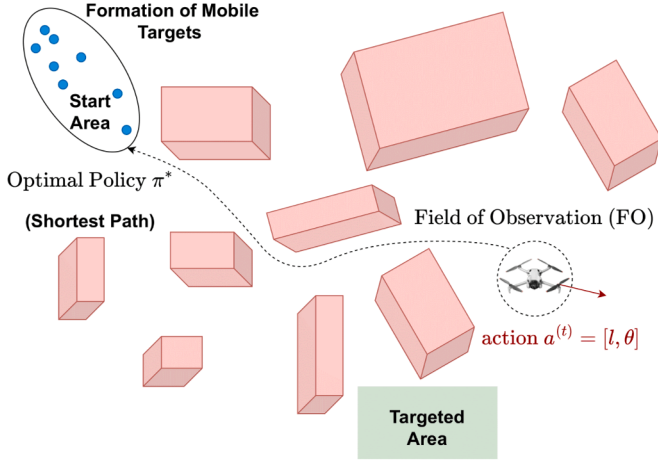


Fig. 1. System Model: Top-View of the UAV Environment. This diagram illustrates the UAV, targets, and obstacles within a bounded, continuous two-dimensional area ($\mathcal{E}$). It also visualizes the UAV's action $\mathbf{a}^{(t)} = [l, \theta]$, the state components $(\mathbf{p}, \mathcal{H}, \mathcal{O})$, and the field of observation (FO). Thus, it provides a visual interpretation of the state and action space models introduced in Eqs.(5) and (4). The UAV's mission is to navigate from a starting area (S_0) to detect all mobile targets ($\mathcal{H}$) while avoiding obstacles ($\mathcal{O}$) and reaching a final zone (S_f).

ded in closed-loop decision making. We address these by unifying offline RL-based navigation, online bandit adaptation, and TWTD-enabled sensing for *multi-target* visitation with dynamic obstacles, thereby coupling mobility, safety, and real-time adaptability.

3. System model & problem formulation

In this section, we describe the system model illustrated in Fig. 1, where a UAV is tasked with searching for mobile targets in a cluttered environment. The UAV's mission is to find the optimal path for visiting all targets while avoiding obstacles.

Notations. We use the following notation conventions for the rest of the paper: scalar values are denoted by lowercase letters (e.g., x). Vectors are represented by bold lowercase letters (e.g., $\mathbf{a}$). Matrices are denoted by bold uppercase letters (e.g., $\mathbf{O}$). Sets are represented by calligraphic uppercase letters (e.g., $\mathcal{M}$). Time-dependent variables are indicated by superscript (e.g., $\mathbf{a}^{(t)}$). Elements of vectors or matrices are indexed using subscripts (e.g., $\mathbf{h}_i$). Functions are denoted by uppercase letters. Furthermore, we refer to a target by its index as a lowercase h , and its location as $\mathbf{h}_i$. We use the $\|\cdot\|$ to represent the Euclidean norm.

3.1. System model

Environment. The environment $\mathcal{E}$ is a continuous and bounded two-dimensional area. Any location is represented as $(x, y) \in \mathbb{R}_+^2 = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq X, 0 \leq y \leq Y\}$, where X and Y define the boundaries. A set of obstacles $\mathcal{O} = \{o_n \mid n \in \{1, \dots, |\mathcal{O}|\}\}$ exists, where each obstacle o_n has a rigid shape but may vary in size and dynamics. The occupied area of an obstacle is defined as $\mathcal{L}_n = \{(x, y) \in \mathbb{R}_+^2 \mid (x, y) \in o_n, \forall n\}$.

Targets. Targets begin in a starting area S_0 and aim to reach a final zone $S_f \in \mathbb{R}_+^2$. The set of mobile targets is $\mathcal{H} = \{\mathbf{h}_i \mid i \in \{1, \dots, |\mathcal{H}|\}\}$, moving in a leader-follower formation. The leader follows waypoints $\mathbf{d}^{(t)}$ toward a goal $\mathbf{e}^{(t)} \in S_f$:

$$\mathbf{d}^{(t+1)} = \mathbf{d}^{(t)} + \alpha(\mathbf{e}^{(t)} - \mathbf{d}^{(t)}), \quad (1)$$

where α controls the leader's step size. Each follower updates its position relative to the leader:

$$\mathbf{h}_i^{(t+1)} = \mathbf{h}_i^{(t)} + \beta(\mathbf{d}^{(t+1)} - \mathbf{h}_i^{(t)}), \quad \mathbf{h}_i \in \mathcal{H}, \quad (2)$$

where β tunes cohesion (larger β keeps followers closer to the leader). To avoid obstacles, target trajectories are smoothed using B-zier curves [37], shaped by control points $C = \{\mathbf{c}_{i,k} \mid i \in \mathcal{H}, k = 0, \dots, e\}$. This yields:

$$B^{(t)}(\mathbf{h}_i) = \sum_{k=0}^e \binom{e}{k} (1-t)^{e-k} t^k \mathbf{c}_{i,k}, \quad \mathbf{h}_i \in \mathcal{H}. \quad (3)$$

B-zier curves ensure smooth paths and allow flexible obstacle avoidance via adjusted control points. If no obstacle exists, targets follow shortest paths from S_0 to S_f . Targets can pass through obstacles (e.g., buildings), necessitating advanced detection such as TWTD.

Action Space. The UAV (agent) flies at fixed altitude and chooses actions $\mathbf{a}^{(t)} = [l^{(t)}, \theta^{(t)}]$, where $l^{(t)} \in [0, L]$ is the step size and $\theta^{(t)} \in [0, 2\pi]$ the direction:

$$\mathcal{A} = \{[l, \theta] \mid l \in [0, L], \theta \in [0, 2\pi]\}. \quad (4)$$

As illustrated in Fig. 1, the action $\mathbf{a}^{(t)} = [l, \theta]$ is directly represented, showing how the UAV selects both step size and movement angle within the environment.

State Space. States $\mathbf{s}^{(t)}$ include UAV position $\mathbf{p}$, target positions $\mathcal{H}$, and obstacle set $\mathcal{O}$:

$$S = \{(\mathbf{p}, \mathcal{H}, \mathcal{O}) \mid \mathbf{p} \in \mathcal{E}, \mathcal{H}, \mathcal{O} \subseteq \mathcal{E}\}. \quad (5)$$

Fig. 1 also depicts these state components, linking the formal definition of S to their visual representation in the environment. The UAV experiences partial observability, relying only on its sensed subset of $\mathbf{s}^{(t)}$. This partial observability is a central challenge, requiring the agent to learn policies that map incomplete observations to effective actions.

Observations. At each step, the UAV perceives targets and obstacles within its field of observation (FO):

$$\mathbf{O}^{(t)} = [\mathbf{p}^{(t)}, t, [D^{(t)}(\mathbf{h}_i)], [O_n^{(t)}]], \quad (6)$$

where $\mathbf{p}^{(t)} = (x^{(t)}, y^{(t)})$, $D^{(t)}(\mathbf{h}_i)$ denotes target detection, and $O_n^{(t)}$ denotes obstacle detection. The FO is a circular area of radius r centered at the UAV. The UAV has no prior map of $\mathcal{E}$.

Sensing Model. The UAV is equipped with a radar-centric sensing suite, with radar serving as the primary detection mechanism. While through-wall detection (TWTD) in its full capacity requires complex pipelines of signal processing, material interaction modeling, and fusion stages [17], we abstract this process to its radar-level output for tractability. This allows us to capture the essential sensing challenges while keeping the model suitable for online analysis.

We consider the preprocessed radar signal matrix $\mathbf{R} \in \mathbb{R}^{M \times N}$, where M and N denote the fast- and slow-time sampling dimensions (range bins and pulses, respectively). Target detection is performed using a cell-averaging constant false alarm rate (CA-CFAR) method [38]. Each cell in $\mathbf{R}$ is compared against a dynamic threshold:

$$U = z \cdot y(q^{-1/e} - 1), \quad (7)$$

where z is a tunable calibration gain, q is the desired false-alarm probability, e is the number of reference cells in the averaging window, and y is the local noise/clutter estimate (the average of the e reference cells). A cell is declared a detection if its magnitude exceeds U .

Propagation Loss and Raw Detection Strength. Detection likelihood is attenuated by propagation losses that increase with range due to two-way path-loss effects [15,39]. Accordingly, the raw detection strength for a hypothesized target location $\mathbf{h}_i \in \mathcal{H}$ at time t is modeled as:

$$P^{(t)}(\mathbf{h}_i) = \frac{A^{(t)}(\mathbf{h}_i)}{U} \cdot \lambda, \quad (8)$$

$$A^{(t)}(\mathbf{h}_i) = k \cdot \|\mathbf{h}_i^{(t)} - \mathbf{p}^{(t)}\|^{-4}, \quad (9)$$

where $A^{(t)}(\mathbf{h}_i)$ captures the received intensity with inverse-quartic distance decay (monostatic two-way path loss), $\lambda \in (0, 1]$ represents attenuation due to occlusions/materials, and $k > 0$ aggregates system constants. Range is inferred from time-of-arrival delay $\tau^{(t)}$ via $d = v \tau^{(t)}$, where v is the propagation speed in the medium.

Processing Gain. We incorporate a processing gain factor to capture improvements from advanced signal processing or fusion methods:

$$G_{\text{proc}} = \frac{\phi_{\text{proc}}}{\phi_{\text{base}}}, \quad (10)$$

where ϕ_{proc} and ϕ_{base} denote the respective post- and pre-processing SNR levels. This gain factor, G_{proc} , provides a high-level representation of cumulative SNR improvements achievable through techniques such as beamforming [40] and matched filtering [41].¹

Final Detection Strength. The overall detection strength for a hypothesized target h_i is given by:

$$D^{(t)}(h_i) = P^{(t)}(h_i) \cdot G_{\text{proc}}, \quad h_i \in \mathcal{H}. \quad (11)$$

The scalar $D^{(t)}(h_i)$ directly governs UAV motion planning within the observable field. Locations outside this field are considered undetectable under the current sensor configuration.

3.2. Problem formulation

The objective is to find an optimal trajectory π^* for the UAV to visit all mobile targets $\mathcal{H}$ at least once while avoiding obstacles $\mathcal{O}$ and satisfying sensing and motion constraints. Let $\mathcal{P} = \{\mathbf{p}^{(0)}, \mathbf{p}^{(1)}, \dots, \mathbf{p}^{(T)}\}$ denote the sequence of UAV positions throughout the mission. Each target $h_i \in \mathcal{H}$ must be detected at least once:

$$\sum_{t=1}^T \delta(\mathbf{p}^{(t)}, \mathbf{h}_i^{(t)}) \geq 1, \quad \forall h_i \in \mathcal{H} \quad (12)$$

where $\delta(\mathbf{p}^{(t)}, \mathbf{h}_i^{(t)}) = 1$ if the target h_i lies within the UAV's field of observation (FO) and has sufficient detection strength at time t , and 0 otherwise. To avoid collisions, the UAV must not enter the occupied area of any obstacle:

$$\mathbf{p}^{(t)} \notin \bigcup_{o=1}^O \mathcal{L}_o, \quad \forall t \in \mathcal{T} \quad (13)$$

The UAV's motion is constrained by a maximum speed v_{max} :

$$\|\mathbf{p}^{(t+1)} - \mathbf{p}^{(t)}\| \leq v_{\text{max}} \cdot \Delta t \quad (14)$$

To confirm target visitation, the detection strength $D^{(t)}(h_i)$ must exceed a minimum threshold D_{min} :

$$D^{(t)}(h_i) \geq D_{\text{min}}, \quad \forall h_i \in \mathcal{H}, \forall t \in \mathcal{T} \quad (15)$$

This ensures that target detections are reliable, accounting for range attenuation and obstacle-induced signal loss, as modeled in the TWTD sensing framework. The full trajectory optimization problem can be expressed as:

$$\mathbf{P}: \min_{\mathcal{P}} \sum_{t=1}^{T-1} \|\mathbf{p}^{(t+1)} - \mathbf{p}^{(t)}\| \quad (16)$$

$$\sum_{t=1}^T \delta(\mathbf{p}^{(t)}, \mathbf{h}_i^{(t)}) \geq 1, \quad \forall h_i \in \mathcal{H} \quad (17a)$$

$$\mathbf{p}^{(t)} \notin \bigcup_{o=1}^O \mathcal{L}_o, \quad \forall t \in \mathcal{T} \quad (17b)$$

$$\|\mathbf{p}^{(t+1)} - \mathbf{p}^{(t)}\| \leq v_{\text{max}} \cdot \Delta t, \quad \forall t \in \mathcal{T} \quad (17c)$$

¹ We emphasize that G_{proc} is not fixed but varies with the specific signal processing pipeline implemented. It is modeled as an abstract parameter capturing the cumulative improvements achievable by different techniques.

$$D^{(t)}(h_i) \geq D_{\text{min}}, \quad \forall h_i \in \mathcal{H}, \forall t \in \mathcal{T} \quad (17d)$$

$$\mathbf{p}^{(t+1)} = \mathbf{p}^{(t)} + \mathbf{a}^{(t)}, \quad \forall t \in \mathcal{T} \quad (17e)$$

Directly solving Problem **P** is intractable in practice, as both the target trajectories $\mathbf{h}_i^{(t)}$ and their corresponding detection strengths $D^{(t)}(h_i)$ are unknown in advance. Therefore, we reformulate this problem using reinforcement learning (RL), where the agent learns to maximize cumulative rewards associated with successful target visits and obstacle avoidance. Through interaction with the environment, the RL agent implicitly learns the transition dynamics and sensing patterns required to approximate the solution to **P**.

3.3. Benchmark path planning algorithm

Before deploying the offline RL solution in online policy selection, we first compare it against a predefined benchmark. Additionally, we evaluate the performance of the online learner against the same benchmark to assess its theoretical guarantees. The objective is to determine the optimal policy π^* that minimizes the total traveled distance while ensuring the UAV successfully visits all unknown targets without colliding with obstacles. Formally, π^* represents the continuous shortest path for the UAV, subject to the constraints defined in **P** (Eq. (16)). To evaluate π^* , we introduce a benchmark algorithm that assumes a perfect information setting, providing an upper bound on achievable performance. While this assumption is not realistic, since targets are initially unknown and must be discovered through exploration, we use it as a reference for comparison to evaluate the gap between our real-world, limited-information approach and an ideal, omniscient one. Under this idealized scenario, the UAV directly optimizes its trajectory by selecting the nearest unvisited target at each step subject to obstacle constraints. If any point $\mathbf{p}^{(t)} \in \mathcal{P}_{\text{dijkstra}}$ intersects with any obstacle, it is removed from the path $\mathbf{p}^{(t)} \cap \mathcal{O} \neq \emptyset \implies \mathbf{p}^{(t)} \notin \mathcal{P}_{\text{dijkstra}}$. The UAV sequentially follows $\mathcal{P}_{\text{dijkstra}}$, updating its position at each step $\mathbf{p}^{(t+1)} = \mathbf{p}^{(t)} + \Delta \mathbf{p}$. The total traveled distance is incrementally computed as:

$$d_{\text{total}} = \sum_{t=0}^T \|\mathbf{p}^{(t+1)} - \mathbf{p}^{(t)}\|. \quad (18)$$

Additionally, the cumulative reward $\hat{\mathbf{R}}$ and final path $\mathcal{P}$ are updated as:

$$\hat{\mathbf{R}} = \sum_{t=0}^T r^{(t)}, \quad \mathcal{P} = \bigcup_{t=0}^T \mathbf{p}^{(t)}. \quad (19)$$

By leveraging perfect knowledge of $\mathcal{H}$, this benchmark provides a lower bound on the UAV's travel distance under full observability. In future work, we will also consider the regulatory and ethical implications of UAV deployment.

4. Multi-target navigation in cluttered environment

To address the real-time multi-target visitation problem in Eq. (16), we use a two-phase framework that separates compute-heavy learning from time-sensitive deployment. The *offline pre-training* stage learns a library of DDPG policies across diverse simulations; the *online deployment* stage selects or adapts a policy from this library during missions (Section 5). This separation enables lightweight decisions in the field while maintaining consistent notation ($\mathbf{s}^{(t)}, \mathbf{O}^{(t)}, \mathbf{a}^{(t)}, \hat{\mathbf{R}}^{(t)}$), as illustrated in Fig. 2.

4.1. Offline training using DDPG model

DDPG [10] is a model-free, off-policy RL method for continuous control with an actor-critic architecture, blending value-based and policy-gradient ideas. Prior work reports effective, collision-free navigation and lower runtime than many classical optimizers [10,11].

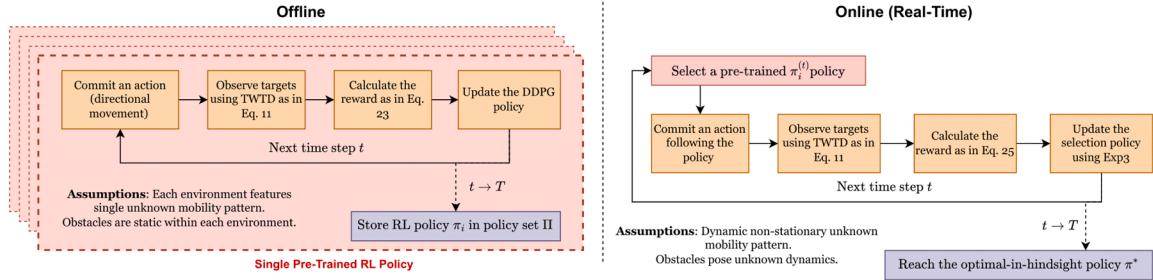


Fig. 2. Hybrid Learning-based Navigation Framework. **Offline:** train a policy set Π with DDPG under varied configurations. **Online:** select $\pi \in \Pi$ to handle unknown, non-stationary environments and approach π^* . Training is shifted offline so the UAV reacts in real time by selecting a pre-trained policy.

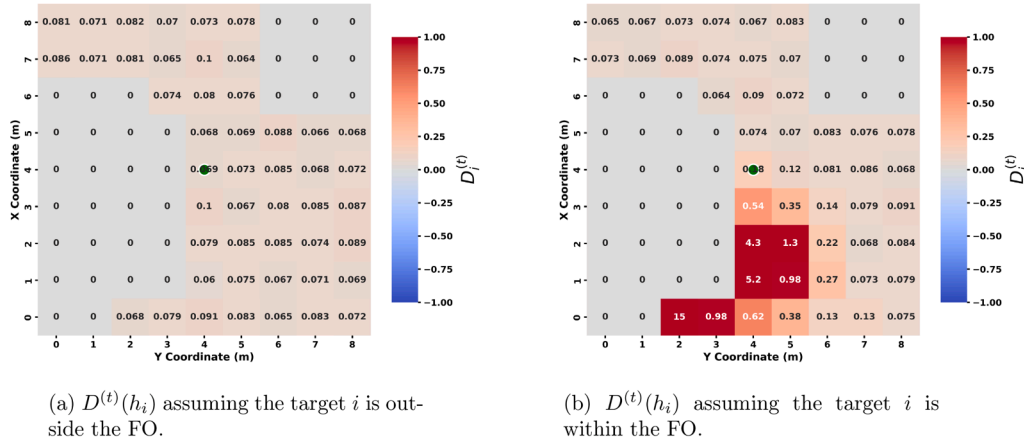


Fig. 3. Example of the Detection strength, given the target(s) at $\mathbf{h}_i^{(t)}$. The figure shows the sensor's output as a matrix. In (a), the target is out of range, resulting in low detection values. In (b), the target is within the FO, yielding higher detection values. The green circle is where the agent is located $\mathbf{p}^{(t)}$ at time t . While the 0s values represent the existence of obstacles at that cell.

DDPG Architecture The actor π_θ maps observation $\mathbf{O}^{(t)}$ to action $\mathbf{a}^{(t)}$, and the critic Q_ϕ evaluates state 0 9 0 0 0 9 action pairs via expected return. The reward $\tilde{R}^{(t)}$ encodes target visitation and obstacle avoidance. See [10] for architectural details.

Training Process Experience replay and target networks stabilize learning. At each step, a transition $(\mathbf{O}^{(t)}, \mathbf{a}^{(t)}, \tilde{R}^{(t)}, \mathbf{O}^{(t+1)})$ is stored; the critic minimizes TD loss and the actor is updated through the critic 0 9 0 0 0 5s gradients. Over episodes this improves the policy toward π^* . We follow Eqs. (4)–(6), and compute rewards only for non-visited targets (let $H_f \subseteq H$ denote the set of *visited* targets).

Reward Function The agent observes detection strength $D^{(t)}(h_i)$ from position $\mathbf{p}^{(t)}$; values are high when targets lie within the FO and near zero otherwise (Fig. 3). To provide a dense, model-based guidance signal consistent with Eq. (16), we define the attraction by penalizing the gap $D^{(t)}(0) - D^{(t)}(h_i)$ and summing only over *undetected* targets $h_i \in H \setminus H_f$ (Eq. (20)). Minimizing this gap increases the expected per-step detection and concentrates learning on targets not yet visited. The shaped reward $\tilde{R}^{(t)}$ then augments $R^{(t)}$ with obstacle/boundary penalties (via χ) and a sensing threshold $D_{\min}$, ensuring safe and efficient trajectories.

To make the attraction signal interpretable, regard $D^{(t)}(0)$ as a nominal *unobstructed, zero-range* detection level (a constant upper reference from the TWTD model). Then

$$R^{(t)}(\mathbf{p}^{(t)}, \{\mathbf{h}_i^{(t)}\}) = \sum_{h_i \in H \setminus H_f} -[D^{(t)}(0) - D^{(t)}(h_i, \mathbf{p}^{(t)} - \mathbf{h}_i^{(t)})]. \quad (20)$$

Safety penalties yield the shaped reward $\tilde{R}^{(t)}$:

$$\tilde{R}^{(t)} = \begin{cases} R^{(t)}(\mathbf{p}^{(t)}, \{\mathbf{h}_i^{(t)}\}), & \mathbf{p}^{(t)} \in \mathcal{E} \setminus \bigcup_{o=1}^{|\mathcal{O}|} \mathcal{L}_o, \exists h_i : D^{(t)}(h_i) \geq D_{\min} \\ -\chi, & \mathbf{p}^{(t)} \in \bigcup_{o=1}^{|\mathcal{O}|} \mathcal{L}_o \\ -2\chi, & \mathbf{p}^{(t)} \text{ near } \partial \mathcal{E} \end{cases} \quad (21)$$

Where χ is a penalty value given to the agent. The value of χ is set to a large number (400) to ensure that the penalty for hitting an obstacle or a boundary is significant enough to compel the agent to learn to avoid them. Furthermore, the agent is penalized if it approaches or intersects the environment boundaries, represented by $x = 0$, $x = X$, $y = 0$, and $y = Y$. This boundary penalty prevents the agent from getting stuck at the edges of $\mathcal{M} \subseteq \mathbb{R}^2$, encouraging exploration. The agent's objective is to reach the optimal policy π , maximizing the reward $\tilde{R}$. The reward function $\tilde{R}$ defines the shortest path for visiting all targets in minimal time while avoiding stationary obstacles $\mathcal{O}$.

Algorithm Algorithm 1 summarizes the training loop: initialize networks, collect transitions with exploration noise, update critic and actor from replay, and iterate over episodes to refine π .

Because real environments are non-stationary, the **online policy selection** phase adapts behavior in real time using the trained library.

5. Policy selection for dynamic obstacle navigation

Real-world UAV navigation faces unpredictable, dynamic obstacles that disrupt fixed mission plans. We propose an online policy-selection framework that leverages a library of pre-trained RL policies, enabling

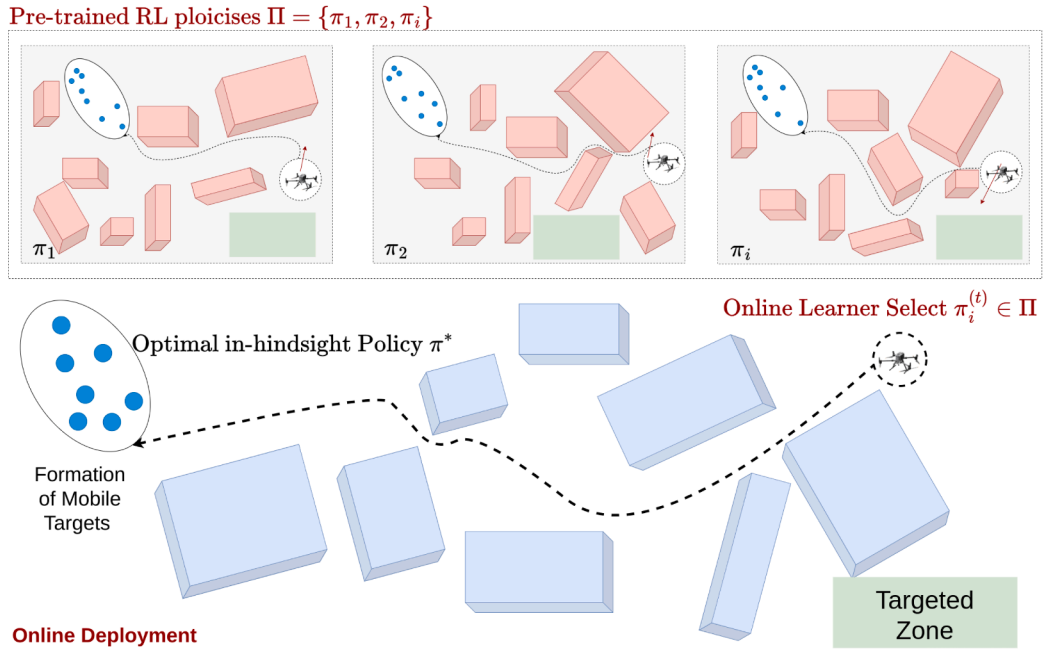


Fig. 4. Pre-trained policies are generated offline by a DDPG model; each policy is trained on distinct environmental configurations. During online deployment, obstacles exhibit unknown, time-varying behavior, reflecting the mobility of targets. The online learner aims to select the optimal in-hindsight policy π^* from a set of pre-trained RL policies Π .

Algorithm 1 Multi-target navigation using DDPG (Offline training phase-pseudo-code).

```

1: Input: Observation space  $\Omega$ , action space  $\mathcal{A}$ , reward  $\tilde{R}^{(t)}$ , actor  $\pi_\theta$ , critic  $Q_\phi$ 
2: Output: Trained policy set  $\Pi = \{\pi_1, \dots, \pi_N\}$ 
3: Initialize actor/critic and replay buffer
4: for each episode do
5:   Observe  $\mathbf{O}^{(0)}$ 
6:   for  $t = 0, \dots, T - 1$  do
7:      $\mathbf{a}^{(t)} = \pi_\theta(\mathbf{O}^{(t)} + \mathcal{N}_t)$ ; execute; observe  $\tilde{R}^{(t)}$ ,  $\mathbf{O}^{(t+1)}$ 
8:     Store transition; update  $Q_\phi$  (TD loss) and  $\pi_\theta$  (policy gradient)
9:   end for
10: end for
11: Return:  $\Pi$ 

```

the UAV to select among them adaptively to maximize cumulative reward under time-varying conditions (Fig. 4).

5.1. Online execution

We assume multiple RL policies are trained offline, each optimized for a distinct environment characterized by obstacle layouts and target motion. Each policy π_b is trained offline on a unique environment $\mathcal{E}_b$ specified by obstacle locations $\mathcal{O}_b$ and target trajectories $\mathcal{H}_b$. To ensure diversity, the environments span dense and sparse obstacle fields, varying target motion patterns, and different levels of clutter, providing the online learner with a flexible library for adaptation. Because obstacles and targets are non-stationary, the mission requires a selector that tracks these changes and approximates the best policy in hindsight. Although we use DDPG for path optimization, other RL algorithms can be substituted as needed. The online learner may also act directly without a fixed policy, following the OCO-based approach in [13].

Pre-Trained RL Policies. Let $\Pi = \{\pi_1, \dots, \pi_N\}$ denote policies trained on environments $\{\mathcal{E}_1, \dots, \mathcal{E}_N\}$, where each $\mathcal{E}_b$ specifies obstacle locations $\mathcal{O}_b$ and target trajectories $\mathcal{H}_b$. Policies are optimized to visit targets while avoiding obstacles (Section 4), and $N = |\Pi|$.

During deployment, an EXP3 (Exponential-weights for Exploration and Exploitation) agent selects a policy from Π at each time step to balance exploration and exploitation under adversarial, non-stationary conditions. EXP3 maintains weights $\{w_n^{(t)}\}$ over the set of policies and updates only the chosen policy according to the observed reward. As rewards are observed, the selector concentrates probability mass on well-performing policies while retaining an exploration floor to remain responsive to shifts.

Identifying the Optimal Policy. The optimal in-hindsight policy $\pi^* \in \Pi$ is

$$\pi^* = \arg \max_{\pi \in \Pi} \sum_{t=1}^T \tilde{R}^{(t)}(\pi), \quad (22)$$

i.e., the single policy that would have maximized cumulative reward over the mission. Although π^* is unknown online, the learner aims to approach its performance.

Regret Metric. We evaluate performance via regret,

$$\text{Reg}(T) = \max_{\pi \in \Pi} \sum_{t=1}^T \tilde{R}^{(t)}(\pi) - \sum_{t=1}^T \tilde{R}^{(t)}(\pi^{(t)}), \quad (23)$$

which measures the gap to π^* . In UAV navigation, low regret means sustaining high detection/visitation rates despite surprise obstacles or target maneuvers narrowing the efficiency gap to an oracle with full prior knowledge.

Regret Bound. With EXP3 and rewards normalized to $[0, 1]$, the regret satisfies (cf. Eq.(11.13) in [42])

$$\text{Reg}(T) \leq 2\sqrt{TN \log N},$$

where T is the horizon and $N = |\Pi|$. Appropriate tuning of the learning rate η yields sublinear regret, ensuring vanishing average regret, i.e., asymptotically optimal average performance.

Policy Selection Algorithm. Algorithm 2 summarizes the deployment procedure. First, it initializes equal weights over the pre-trained policies. Then, at each round, it samples a policy from the exponential-weights mixture, executes the chosen policy in the environment, and updates its weight using an importance-weighted reward. This loop

balances exploration with exploitation and adapts naturally to non-stationarity. As shown in Fig. 2, Algorithm 2 operates directly on the pre-trained policies generated by Algorithm 1, treating them as arms in the online selection problem. In particular, Algorithm 2 samples policies using a softmax-style distribution induced by EXP3's exponential weights (line4), and the resulting multiplicative-weights updates yield adversarial no-regret guarantees [9,42].

Non-stationarity appears as: (i) sudden obstacle insertions collision spikes down-weight sparse field policies and shift mass to dense-field ones; (ii) target accelerations missed visits increase weight on fast kinematics policies; (iii) periodic switching the exploration floor prevents forgetting, enabling rapid reactivation of previously down-weighted policies (Fig. 4).

Algorithm 2 Online policy selection with EXP3 deployment phase.

```

1: Input: Policies  $\Pi = \{\pi_1, \dots, \pi_N\}$ , learning rate  $\eta$ , exploration rate  $\gamma$ .
2: Initialize:  $w_n^{(0)} \leftarrow 1$  for  $n = 1, \dots, N$ .
3: for  $t = 1, \dots, T$  do
4:   Form  $P^{(t)}(\pi_n) = (1 - \gamma) \frac{w_n^{(t)}}{\sum_{j=1}^N w_j^{(t)}} + \frac{\gamma}{N}$ .
5:   Sample index  $n_t \sim P^{(t)}$ ; execute  $\pi_{n_t}$ ; observe reward  $\tilde{R}^{(t)}(\pi_{n_t})$ .
6:   Set  $\hat{R}^{(t)} \leftarrow \tilde{R}^{(t)}(\pi_{n_t}) / P^{(t)}(\pi_{n_t})$ .
7:   Update  $w_{n_t}^{(t+1)} \leftarrow w_{n_t}^{(t)} \exp(\eta \hat{R}^{(t)})$ .
8: end for

```

5.2. Mini-batched EXP3 with switching cost

Frequent controller handovers can induce transients and *thrashing*. We stabilize adaptation with *mini-batching* and an explicit *switch cost*. The learner commits to a policy for τ steps, then updates once per batch using the average reward; switches across batch boundaries incur a penalty μ :

$$\hat{R}^{(t)}(\pi^{(t)}) = \tilde{R}^{(t)}(\pi^{(t)}) - \mu \mathbb{1}\{\pi^{(t)} \neq \pi^{(t-1)}\}, \quad \mu \geq 0, \quad (24)$$

$$\bar{R}_j(\pi) = \frac{1}{\tau} \sum_{k=1}^{\tau} \tilde{R}^{(t_j+k)}(\pi), \quad t_j = (j-1)\tau. \quad (25)$$

Let $R(T')$ denote a regret bound for standard EXP3 over T' rounds. Wrapping it in τ -step batches yields the *policy-regret* bound:

$$\text{Reg}(T) \leq \tau R\left(\frac{T}{\tau}\right) + O\left(\frac{T}{\tau} + \tau\right),$$

which motivates choosing τ to balance *agility* (small τ) and *stability* (large τ). A practical initialization is

$$\tau^* \propto (7N \log N)^{-\frac{1}{3}} T^{\frac{1}{3}},$$

refined from flight logs. Algorithm 3 lists the procedure.

Algorithm 3 EXP3- τ : EXP3 with switching cost pseudo-code.

```

1: Input:  $\Pi, \eta, \tau, \mu, T$ 
2: Init:  $w_n \leftarrow 1$ ;  $\pi_{\text{prev}} \leftarrow \text{None}$ ;  $J \leftarrow \lfloor T/\tau \rfloor$ 
3: for  $j = 1$  to  $J$  do
4:   Form  $p_n^{(j)}$  from  $\{w_n\}$  (with smoothing); sample  $\pi^{(j)}$ 
5:   Execute  $\pi^{(j)}$  for  $\tau$  steps
6:   Compute  $\bar{R}_j$  (penalize only the batch-boundary switch if  $\pi^{(j)} \neq \pi_{\text{prev}}$ )
7:    $w_{\pi^{(j)}} \leftarrow w_{\pi^{(j)}} \exp(\eta \bar{R}_j / p_{\pi^{(j)}}^{(j)})$ 
8:    $\pi_{\text{prev}} \leftarrow \pi^{(j)}$ 
9: end for

```

Tuning in practice. Larger τ reduces switches and smooths trajectories; smaller τ reacts faster to new obstacles. Set μ from logs to match the observed transient cost of switching controllers. We report $\text{Reg}(T)$, total switches, collision rate, and visitation latency to quantify the stability Cagility trade-off.

6. Performance evaluation

To evaluate the proposed hybrid solution, we present the results in a reader-first order: (A) DDPG training ablations, (B) train-time sensitivity to processing gain, (C) generalization under multi-task RL, and (D) online policy selection under dynamic obstacles. We first summarize the environment, TWTD sensing, and DDPG configuration; then we define evaluation metrics; finally, we report and discuss results with consistent indicators across experiments. We examine: (i) the agent 0 9 0 0 0 5s response to different numbers of targets and clutter occlusions; (ii) the effect of processing gain during DDPG training; (iii) generalization gaps under multi-task training; and (iv) the online learner 0 9 0 0 0 5s behavior under dynamic, non-stationary obstacles. We first describe the environment, TWTD sensing, and the DDPG model; we then introduce the evaluation metrics, and finally present the results.

Experimental Setup. We specify key parameters for three components: the environment, the TWTD sensing model, and the Deep Deterministic Policy Gradient (DDPG) agent. Table 1 reports all hyperparameters. Our analysis focuses on navigation behavior; communication-layer effects (delays, interference, packet loss) are out of scope here and left to future work. The top row of Fig. 5 shows the cumulative rewards achieved during training, indicating how learning dynamics evolve and stabilize across the experimental conditions.

Environment. The environment size $\mathcal{E}$ is $\mathcal{E}(X, Y) = (0, 55)$. The action space is $\mathcal{A} = \{[l, \theta] \mid l \in [0, 5], \theta \in [0, 2\pi)\} \subseteq \mathbb{R}^2$, where 5 is the maximum step size. We set the minimum detection strength $D_{\min} = 2$, which marks the i th target as *visited*. A mission is considered a success when all targets detected/visited within the episode horizon, with zero collisions and no boundary violation. If the threshold is not met, the agent is rewarded according to Eq. (20). The penalty χ for hitting an obstacle or boundary is 400. The step sizes α and β controlling leader and follower motion are both 0.2. The starting area is $S_0 = \{(x, y) \mid x \in [0, 6], y \in [0, 6]\}$. The code for reproducing the experiments can be found in GitHub.²

TWTD. The number of training cells w_{train} used in the CFAR algorithm is set to 10, and the false alarm rate q is 10^{-4} . These parameters influence target detection sensitivity, with the attenuation factor λ set to 0.4, the processing parameter ϕ_{proc} to 5, and the base peak factor ϕ_{base} to 3. The leader's endpoint $\mathbf{e}$ follows a Gaussian distribution $\mathcal{N}((55, 55), 0.8^2)$.

Evaluation Metrics. We measure performance during training, inference, and online adaptation.

Training. We track convergence of cumulative reward and loss. We interpret convergence relative to a benchmark that assumes perfect sensing and shortest-path routing, gauging how quickly the DDPG agent approaches near-optimal behavior.

Inference. We report *success rate* (number of targets visited), *Reward*, *Distance* (path length), *Steps* (time to complete), *Obstacle Hit Ratio* (safety proxy). An ideal Obstacle Hit Ratio is 0, indicating collision-free navigation; larger values indicate reduced safety. We compare against the benchmark to contextualize efficiency and safety.

Online adaptation. We use *Average Regret* and *Policy Probability* traces to assess how quickly the selector identifies and exploits the best policy as conditions change. Regret is defined as the cumulative performance gap between the chosen and the best policy over time; smaller regret indicates faster adaptation under dynamic, potentially adversarial drift.

Our evaluation integrates results across all modules. We first study DDPG under increasing clutter, target counts, and processing gain; next we show limited generalizability under multi-task RL; finally we present EXP3 vs. UCB results, including ablations on LOS vs. TWTD and processing gain during online deployment. This layout isolates the effect of each pipeline component and uses consistent indicators for cross-analysis.

² <https://github.com/Noor-Khial/DRONE.git>

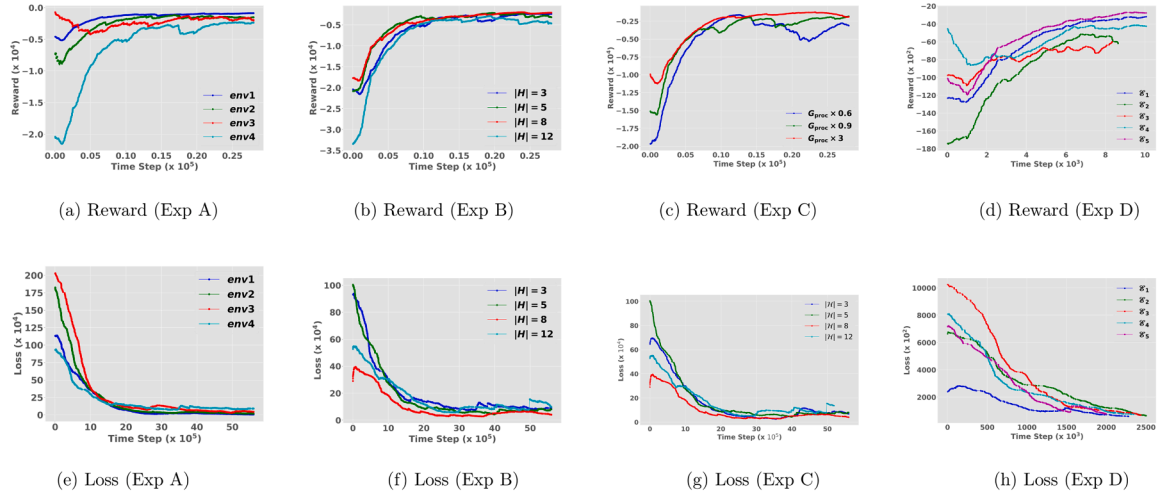


Fig. 5. Training results of the DDPG agent across four experimental setups. (1) **Exp A:** Reward and loss under varying clutter levels. The reward curves evidence steady learning across clutter regimes: as obstacles increase, the agent takes slightly longer to improve but converges to safer paths with fewer boundary interactions. (2) **Exp B:** Reward and loss as a function of processing strength. (3) **Exp C:** Reward and loss under an alternative processing-strength configuration. (4) **Exp D:** Reward and loss when training across multiple $\mathcal{E}$, each with a distinct obstacle configuration. Across experiments, the DDPG agent learns policies that remain stable across heterogeneous layouts and sensing settings, supporting later online selection.

Table 1
Parameters for the environment and models.

Environment		TWTD		DDPG Model	
Parameter	Value	Parameter	Value	Parameter	Value
$\mathcal{E}(X, Y)$	(0, 55)	w_{train}	10	Critic learning rate	10^{-3}
θ	$[0, 2\pi]$	q	10^{-4}	Actor learning rate	10^{-4}
L	$[0, 5]$	λ	0.4	γ	0.99
S_0	$\{(x, y) \mid x \in [0, 6], y \in [0, 6]\}$	ϕ_{proc}	5	τ	0.005
$D_{\min}$	2	ϕ_{base}	3	Buffer size	20000
χ	400	k	0.5	Batch size	64
α, β	0.2	$\mathbf{e}$	$\mathcal{N}((55, 55), 0.8^2)$	Number of Time Steps	200

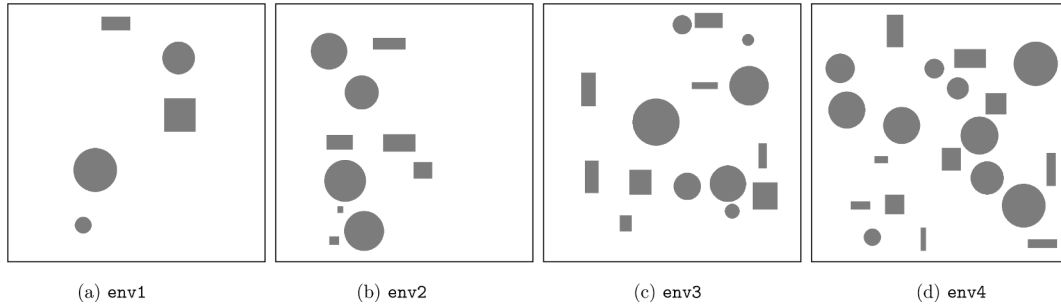


Fig. 6. Four environments (env1 Cenv4) illustrate different clutter levels for UAV navigation. Black boxes denote obstacles; more boxes indicate higher clutter and increased navigation difficulty.

6.1. Impact of clutter level on the agent's performance

We evaluate how obstacle clutter affects performance using four environments with increasing obstacle density (Fig. 6).

Discussion. Fig. 5a shows that more clutter increases mission difficulty. Table 2 indicates that the DDPG agent maintains high success across environments (e.g., 95% in env4) but follows longer paths than the benchmark (e.g., 43.62 vs. 25.14 in env4) as it prioritizes obstacle avoidance. This strategy reduces steps (e.g., 14 vs. 19) and yields a less negative cumulative reward (−802.49 vs. −1496.07), highlighting a safety–efficiency trade-off. Overall, the agent emphasizes collision-free navigation while preserving strong success rates. This is a crucial characteristic for real-world UAV applications where avoiding collisions is paramount. The computational time for policy selection during online

deployment is negligible compared to the agent's action execution. Both the EXP3 and UCB algorithms are computationally lightweight and well suited for real-time decision making, which is a crucial characteristic for online UAV applications.

6.2. Impact of number of targets on the performance of the agent

We study scalability by varying the number of mobile targets in env3 (Fig. 6c). Targets originate from S_0 and each end point $\mathbf{e}^{(i)}$ is sampled from the normal distribution in Table 1. We evaluate 3, 5, 8, and 12 targets.

Discussion. As shown in Table 3, increasing the number of targets make the task harder: rewards become more negative and training loss rises (Fig. 5b and f). During inference, the DDPG agent's success rate

Table 2

Performance of the DDPG agent during inference with varying clutter. **Note:** Bold marks per-metric best. $\uparrow$ higher-is-better; $\downarrow$ lower-is-better.

Env	Success %		Path Dist.		Steps		Obstacle Hit %		Cumulative Reward	
	DDPG	Dijkstra	DDPG	Dijkstra	DDPG	Dijkstra	DDPG	Dijkstra	DDPG	Dijkstra
env1	100	100	26.01	24.56	6	17	0	0	-610.69	-1253.81
env2	95	100	32.41	26.97	7	18	0	0	-630.65	-1256.64
env3	95	100	31.63	26.56	9	15	0	0	-649.60	-1221.00
env4	95	100	43.62	25.14	14	19	0	0	-802.49	-1496.07

Table 3

Inference performance of the DDPG agent under varying numbers of targets. **Note:** Bold marks per-metric best. $\uparrow$ higher-is-better; $\downarrow$ lower-is-better.

# Targets	Success %		Path Dist.		Steps		Obstacle Hit %		Cumulative Reward	
	DDPG	Dijkstra	DDPG	Dijkstra	DDPG	Dijkstra	DDPG	Dijkstra	DDPG	Dijkstra
3	95	100	30.6	23.5	19	19	0.1	0	-1562.4	-1493.5
5	90	100	42.2	24.6	20	20	0.2	0	-1830.3	-3562.5
8	90	100	44.7	27.3	24	25	0.3	0	-2079.1	-6147.7
12	86.7	100	45.8	29.3	29	27	0.37	0	-2356.1	-6906.9

Table 4

DDPG inference performance under varying processing strength. **Note:** Bold = best performance. $\uparrow$: higher is better (Success, Reward); $\downarrow$: lower is better (Distance, Steps, Hits).

G_{proc}	Success Rate % $\uparrow$		Path Dist. $\downarrow$		Steps $\downarrow$		Obstacle Hit % $\downarrow$		Cumulative Reward $\uparrow$	
	DDPG	Dijkstra	DDPG	Dijkstra	DDPG	Dijkstra	DDPG	Dijkstra	DDPG	Dijkstra
$\mathcal{T}_{s0.6}$	86.7	100	42.2	25.38	29	19	0.1	0	-1562.4	-1493.5
$\mathcal{T}_{s0.9}$	90	100	33.6	25.38	20	19	0	0	-1395.9	-1492.4
$\mathcal{T}_{s3}$	90	100	33.4	25.38	15	19	0	0	-1043	-1490.3

decreases moderately (from 95 % with 3 targets to 86.7 % with 12), whereas the benchmark remains at 100 %. Path distance grows with more targets (e.g., 30.60 to 45.89 units from 3 to 12 targets), and obstacle-hit events rise (from 0.10 to 0.37) while the benchmark records none. Overall, added targets induce longer routes and more time steps, stressing the agent's planning and avoidance. The scalability of the framework to significantly larger environments and a greater number of targets is therefore a key area for future research. Although our experiments demonstrate how performance degrades with increasing targets, further investigation, including multiagent extensions, will be necessary to validate effectiveness in expansive and complex real-world scenarios.

6.3. Impact of processing gain G_{proc} during DDPG training

We study how the DDPG agent behaves under changes to processing gain during *training* to separate train-time sensitivity from online adaptation. We vary the processing strength G_{proc} to study how the relation between the base signal ϕ_{base} and the processed signal ϕ_{proc} affects training and inference. We test $G_{proc} \in \{\mathcal{T}_{s0.6}, \mathcal{T}_{s0.9}, \mathcal{T}_{s3}\}$.

Discussion. Increasing G_{proc} Improving the processing strength G_{proc} improves the DDPG agent's performance, as shown in Fig. 5c and g, and Table 4. At lower G_{proc} values, such as 0.6, the agent struggles with target detection, leading to suboptimal performance with higher total distance and more time steps. As G_{proc} increases to 0.9 and 3, the agent's performance improves, reducing both distance traveled and time steps, and aligning more closely with the optimal policy (benchmark). These results demonstrate that higher processing strength enhances the agent's efficiency in completing the mission.

6.4. DDPG limited generalizability (RL with multi-task)

We assess generalization under non-stationarity by evaluating a single DDPG policy over multiple shifted environments. According to Table 5, performance degrades as the environment shifts, indicating the

Table 5

Inference performance under increasing environment shifts (non-stationarity). **Note:** In the tables, bolded values represent the best performance for the DDPG agent in that specific metric.

Environment	DDPG	Dijkstra
Shift by 20 % ($\mathcal{E}_1$)	-1111.90	-1047.44
Shift by 40 % ($\mathcal{E}_2$)	-2886.50	-1496.04
Shift by 80 % ($\mathcal{E}_4$)	-4361.51	-1278.22
Shift by 100 % ($\mathcal{E}_5$)	-10016.19	-1745.56

need for adaptation. This observation underscores the primary motivation for our online selection module: to maintain performance under unforeseen changes without the need for time-consuming re-training Fig. 7.

6.5. Performance of the online learner

We evaluate the proposed online learner across sensing, algorithmic, and environmental dimensions. Concretely, we study: (i) sensing modality Through-Wall Target Detection (TWTd) versus line-of-sight (LOS); (ii) sensor processing gain G_{proc} ; (iii) exploration tuning and selector design (EXP3, UCB, and EXP3 with switching-cost control); and (iv) robustness under switching and abrupt policy changes. We report regret, success rate, obstacle-collision rate, and cumulative reward, providing a holistic view of online decision quality. The online deployment phase has a cost of $O(N)$ [42].

6.5.1. Tuning the EXP3 learning rate η

The learning rate η governs the exponential reweighting in EXP3, trading off stability against speed of adaptation. Smaller η yields conservative updates and fewer collisions but slower convergence; larger η accelerates learning at the risk of instability. We sweep four η values and measure mean regret, success, collisions, and cumulative reward.

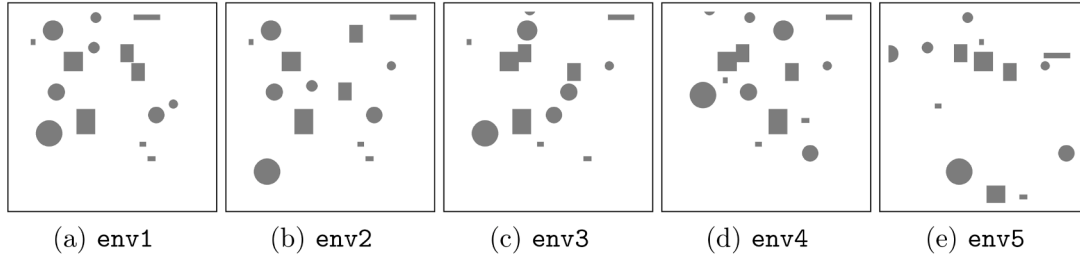


Fig. 7. Five environments (env1–env5) with increasing shifts from the training distribution, modeling different levels of non-stationarity by shifting obstacle configurations. These environments are used to pre-train diverse policies (Π), supporting online adaptation under unseen conditions.

Table 6

EXP3 tuning: learning rate η .

η	Average Regret ↓	Success % ↑	Obstacle Hit % ↓	Cumulative Reward ↑
0.05	0.42	93	1.76	−801.96
0.10	0.05	100	6	−8700.78
0.15	0.25	100	13.77	−15080.32
0.20	0.20	100	19.03	−12666.68

Table 7

TWTD vs. LOS.

Mode	Average Regret ↓	Success % ↑	Obstacle Hit % ↓	Cumulative Reward ↑
TWTD	0.07	93	10.14	−4277.34
LOS	0.02	95.3	19.20	−49855.01

Table 8

Effect of processing gain G_{proc} .

G_{proc}	Average Regret ↓	Success % ↑	Obstacle Hit % ↓	Cumulative Reward ↑
0.6	0.03	92.7	7.33	−1577.29
0.8	0.03	100	16.48	−15432.64
1.0	0.25	100	23.64	−4839.29
1.2	0.15	100	22.44	−4141.19

Discussion. Table 6 shows that $\eta = 0.10$ strikes the best balance: regret is minimal with stable navigation. Very small η (0.05) slows adaptation (higher regret) but suppresses collisions; larger η (≥ 0.15) overreacts to short-term rewards, increasing collisions and degrading cumulative reward. We therefore use $\eta \approx 0.10$ as a robust default for dynamic UAV settings.

6.5.2. LOS vs. TWTD sensing

We compare TWTD-enabled sensing against LOS-only sensing, where LOS is modeled as a binary reward (1 on target detection, 0 otherwise) [13,36].

Discussion. As shown in Table 7, TWTD lowers collision rates relative to LOS, indicating safer navigation in clutter where LOS is visibility-limited. Success rates are comparable, but TWTD reduces cumulative penalties by improving spatial awareness and path choice. LOS-only sensing tends to overcommit to visible but risky routes, increasing collisions and cost.

6.5.3. Effect of processing gain G_{proc}

G_{proc} models signal-processing amplification, improving detection confidence (and potentially exploration) at higher compute cost. We sweep G_{proc} and evaluate performance.

Discussion. Moderate gains stabilize detection while avoiding excessive exploration Table 8; the range $G_{\text{proc}} \in [0.8, 1.0]$ balances success and safety. Very high gains saturate detection benefits yet invite more aggressive moves (increasing collisions) without commensurate reward improvements.

6.5.4. Policy selection: EXP3 vs. UCB and policy-set size

We assess the online selector that mixes among pre-trained policies Π using either EXP3 or UCB [33] in non-stationary obstacle fields. Larger $|\Pi|$ generally increases exploration burden. Fig. 9 shows representative distributions A–D for which the best policy changes over time. Fig. 8 shows the UAV trajectory at time slots $t \in \{3, 10, 20\}$, and the corresponding executed policy is shown in Fig. 9A.

Discussion. Across A–B (Fig. 9a–b), EXP3 concentrates probability mass faster and achieves lower regret than UCB. The regret penalty increases with $|\Pi|$ due to additional exploration, yet EXP3 remains more responsive in non-stationary regimes.

6.5.5. EXP3 with switching cost

To temper over-switching, we introduce a time-windowed variant EXP3- τ that penalizes frequent policy changes via a window parameter τ .

Discussion. EXP3- τ lowers regret and reaches 100 % success Table 9, confirming that temporal consistency stabilizes learning under fluctuations. Larger τ reduces regret further but raises collision rates (stickier behavior delays pivots). $\tau = 5$ offers a practical balance.

6.5.6. Comparison with meta-RL

We benchmark against a first-order MAML (FOMAML) agent with a categorical policy (2-layer MLP, 128 ReLU units). A fast-adapted copy performs one inner SGD step on a 1-step horizon; its first-order policy gradient updates the base policy via Adam. Returns are done-masked Monte Carlo with $\gamma = 0.99$, standardized, and include an entropy bonus $\beta = 0.01$ to promote exploration Tables 10.

Discussion. Meta-RL attains competitive success but incurs higher regret due to slower within-episode adaptation when layouts drift. EXP3 rapidly reweights among fixed policies, cutting regret but increasing collisions under our penalty structure. EXP3- τ further reduces regret toward zero by discouraging frequent switches; the induced stickiness delays pivots after shifts, increasing transient collisions. Hence, EXP3/EXP3- τ dominate when *time-to-adapt* and regret are primary; meta-RL can be preferable when *safer trajectories* under dense clutter are paramount and dynamics are near-stationary.

6.5.7. Abrupt policy change: EXP3 vs. UCB

We simulate an abrupt shift of the optimal policy from distribution C to D: C over $T \in [0, 50]$ and D over $T \in [50, 200]$.

Discussion. EXP3 maintains low regret and quickly reallocates probability after the C→D switch (Fig. 10c–d). UCB adapts more slowly, accruing higher transient regret during the shift, underscoring EXP3's responsiveness to abrupt changes.

6.5.8. EXP3 under levels of non-stationary detection

We parameterize non-stationarity by an adversarial level $a \in [0, 1]$ that perturbs: (i) obstacles (spawn/shift/inflate), (ii) target motion (velocity and waypoint resampling), and (iii) sensing (occlusions and detection-strength skew). $a = 0.0$ is benign i.i.d.; $a = 0.3$ injects mild intermittent shifts; $a = 0.6$ induces frequent shifts partially coupled to the

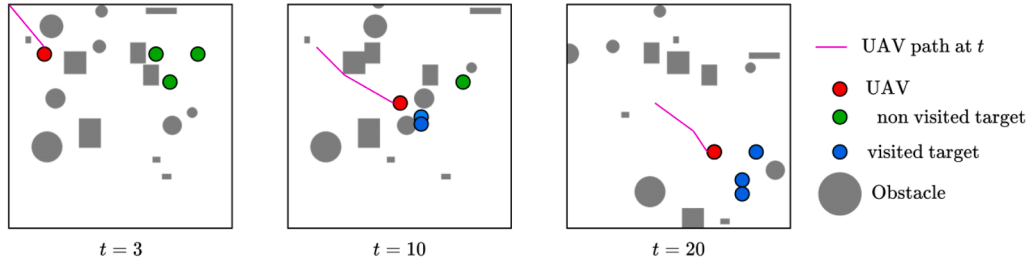


Fig. 8. UAV trajectory during online deployment. Each sub-figure presents a snapshot of the environment at time t , illustrating the UAV's trajectory generated by the currently executed pre-trained policy corresponding to the distribution shown in Fig. 9A.

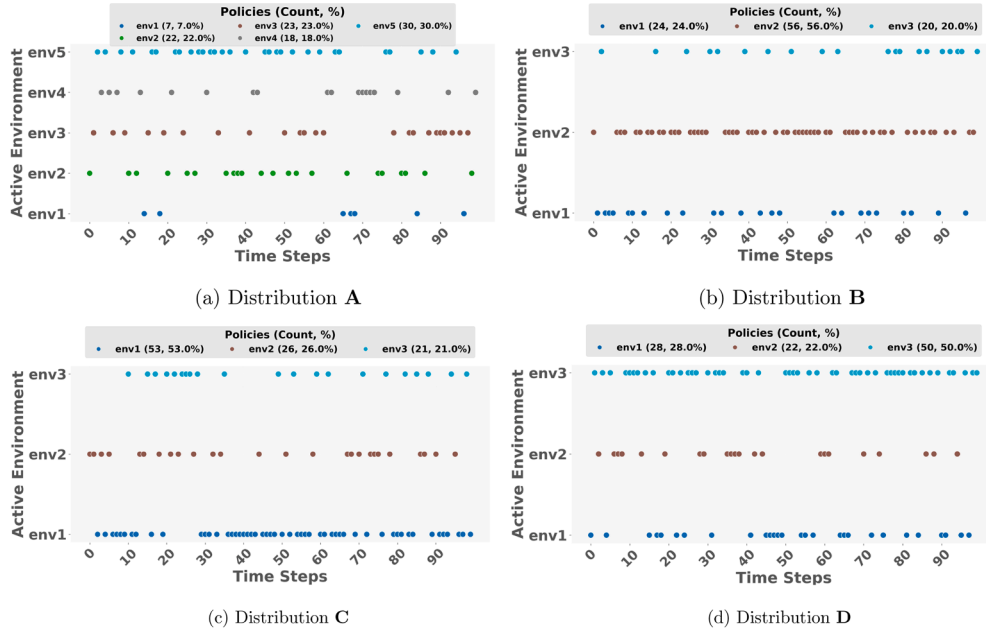


Fig. 9. Non-stationary environment distributions (A–D) showing how the active policy changes over time. Points mark policies selected at each step, highlighting switching behavior under dynamic obstacle configurations.

Table 9
Online learner with switching cost.

Algorithm (τ)	Average Regret $\downarrow$	Success % $\uparrow$	Obstacle Hit % $\downarrow$	Cumulative Reward $\uparrow$
EXP3 (τ)	0.004	92.7	5.10	−2258.17
EXP3- τ (3)	0.0025	100	9.03	−11071
EXP3- τ (5)	0.0008	100	11.62	−9692.78
EXP3- τ (10)	0.0011	100	17.88	−6551.44

Table 10
Comparison with state-of-the-art.

Algorithm	Mean Regret $\downarrow$	Targets Found $\uparrow$	Collisions $\downarrow$	Cumulative Reward $\uparrow$
Meta-RL (FOMAML)	0.25	87	8.86	−12724
EXP3	0.03	100	18.09	−10381
EXP3- τ (10)	0.001	100	28.01	−22838

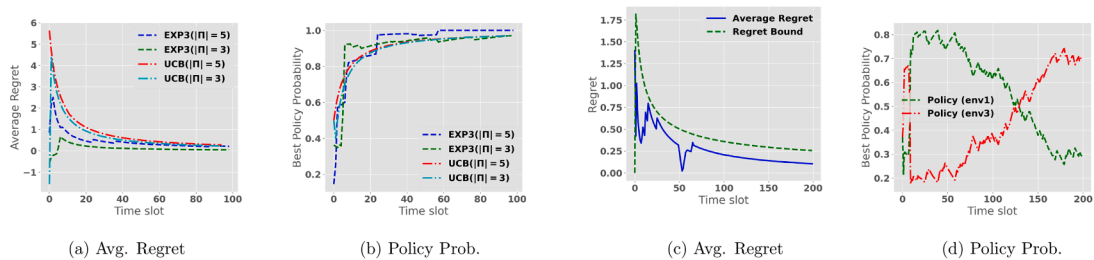


Fig. 10. EXP3 vs. UCB across two non-stationary regimes. (a)–(b): dynamic obstacles; (c)–(d): abrupt policy-change regime (C–D). Fig. 10 c illustrates the bounded-regret response.

Table 11
Performance vs. adversarial level.

Adversarial Level	Average Regret ↓	Success % ↑	Obstacle Hit % ↓	Cumulative Reward ↑
0.0	0.02	92	9.9	−3630
0.3	0.03	100	12.95	−32446
0.6	0.11	100	18.9	−38383
1.0	0.85	100	25.35	−34656

agent; $a=1.0$ yields fully adaptive worst-case reconfigurations aligned against the current dominant policy. We report mean regret, success, collisions, and cumulative reward (Table 11).

Discussion. Regret and collisions rise monotonically with a , while success saturates near 100 % for $a \geq 0.3$. Thus, EXP3 reliably finds targets even under strong shifts, but adversarial occlusions and reconfigurations lengthen routes and increase penalty-weighted costs. The jump from $a=0.6$ to $a=1.0$ reflects fully adaptive shifts that transiently mislead the selector before weights reconcentrate. At $a \leq 0.3$, regret ≤ 0.03 matches near-stationary performance of the best fixed policy.

Overall Takeaways. EXP3-based selection consistently outperforms UCB and heuristics in non-stationary settings, maintaining low regret and fast adaptation. TWTD sensing and moderate G_{proc} improve detection reliability and navigation safety. Adding a switching-cost window (EXP3- τ) introduces temporal smoothness that further reduces regret at the expense of delayed pivots. These trends support deployment in dynamic UAV missions where sensing, mobility, and learning must co-adapt to unpredictable changes.

7. Conclusion

In this paper, we proposed a hybrid learning-based framework that integrates pre-trained RL policies with an OCO-driven online policy selection mechanism for multi-target visitation in cluttered environments. The framework accounts for non-stationary dynamics in both target mobility and obstacle configurations. Experimental results show that increasing the number of targets raises mission duration, while higher clutter density prolongs training. The online learner exhibits superior adaptability to varying non-stationarity levels compared to UCB, underscoring the advantage of regret-based policy selection in dynamic UAV missions.

8. Limitations and future work

This study presents a simulation-driven framework that couples learned policies with online policy selection for multi-target visitation under uncertainty and non-stationarity. While the results highlight benefits in adaptability and safety, several limitations point to natural extensions.

Communication robustness. Our evaluation assumes reliable feedback, whereas real deployments face delays, packet loss, and dropouts. Future work could consider delay-aware selection mechanisms that discount observations and use short-horizon buffered control.

From single- to multi-UAV coordination. We focus on a single UAV in cluttered environments. Scaling to teams introduces interference, shared airspace, and heterogeneous platforms. A future direction is to study decentralized selection where vehicles coordinate via lightweight information sharing or sparse messaging.

Sensing fidelity and noise. TWTD is beneficial in clutter, but its performance under low SNR and structured false alarms is not fully characterized. We will conduct SNR sweeps and controlled false-alarm studies to analyze precision recall, detour costs, and regret, and to tune processing gain schedules.

Adversarial and extreme environments. Our non-stationarity regime can be extended to abrupt obstacle insertions and highly cluttered maps. Combining time-windowed selection with change-point de-

tectors is a promising direction to reduce adaptation latency and quantify recovery after distributional shifts.

System resilience. We do not explicitly model failures such as GPS drift, intermittent sensors, or actuator faults. Future work will evaluate conservative fallback policies (e.g., safe loiter and re-localization) and quantify failure containment, stabilization time, and regret overshoots during recovery.

CRedit authorship contribution statement

Noor Khial: Writing – original draft, Methodology, Formal analysis, Conceptualization; **Mhd Saria Allahham:** Writing – review & editing, Supervision, Methodology, Conceptualization; **Naram Mhaisen:** Writing – review & editing, Supervision, Methodology, Conceptualization; **Loay Ismail:** Writing – review & editing, Supervision, Conceptualization; **Mohamed Mabrok:** Supervision, Methodology, Conceptualization; **Amr Mohamed:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Noor Khial reports financial support was provided by Qatar University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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